

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 15

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• All greenhouse gases expressed as CO₂ equivalent (allowing for different warming strengths) • Direct emissions from fuel combustion plus indirect: manufacture of components, maintenance items, final disposal (life-cycle analysis) • A fair share of the infrastructure: roads, car parks, signals, street lighting	3
01.2	• Compares the populations of over 18 000 populations of 3700 vertebrate species (terrestrial, freshwater, marine) with 1970 levels • Published in the Living Planet Report as a measure of the state of the planet's ecosystems / biodiversity loss over time	2
01.3	• Hardy, disease-resistant Boran cattle; regular tick dipping reduces ticks for wild herbivores too • Mobile bomas (night enclosures) protect cattle from lions and leave dung that enriches soil and plant growth • Electric fences keep game in and poachers out; biological corridors let animals (except rhinos) move to other reserves; problem elephants relocated • Monitoring: radio-collared lions, drones, camera traps, track surveys; endangered species (Grevy's zebra, northern white rhino) protected • Community links: ecotourism income funds health, education and water projects; local employment; cattle breeding help for local farmers	5
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 380-383; AQA spec 3.6.4.2.	
02.1	• Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so	2

Question	Marking guidance	Marks
02.2	• Total impact = population size x the individual (per capita) impact of each person • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact	2
02.3	<i>Any 1 from:</i> • Slash and burn agriculture • Fishing • Discharge of sewage / biodegradable pollutants	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
03.1	• A community maintained at a stage before the climax by regular human activity (grazing, mowing, burning, coppicing) that deflects succession • Chalk grassland was produced by centuries of sheep grazing; without it succession to scrub and woodland resumes	2
03.2	• Grazing removes taller / dominant grasses and prevents scrub establishing (5% cover) so small plants are not shaded out • Trampling creates bare ground for wildflower seed germination and warm patches for invertebrates / reptiles • Dung creates small nutrient-rich patches supporting different plants and invertebrates; sheep eat short grasses and some bushes • Mowing removes everything at once including plants grazers avoid, and removes flowers / seeds before setting; burning removes vegetation but also kills invertebrates and allows scrub regrowth; abandoned plot dominated by scrub (55%) so few species	4
03.3	• Sheep eat short grass (some breeds eat scrub); ponies avoid flowers but eat bracken; cattle remove long rough grass • Choice depends on the vegetation to be controlled and the plants to be protected	2
03.4	• Removes mature heather bushes and young trees that grazers will not eat, creating open unshaded ground • Heat stimulates dormant heather seeds to germinate so young heather re-establishes	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 47-49; AQA spec 3.1.3.3.	
04.1	• A: coppice rotation creates a patchwork of different ages / light levels - clearings with wildflowers for butterflies, dense regrowth for nesting birds; oak supports many invertebrates • B: same tree species but the closed canopy shades the floor so few flowering plants, fewer invertebrates and butterflies; fewer niches • C: non-indigenous single-species, single-age conifers support few native invertebrates; dense shade; no clearings or dead wood; needles acidify soil • Habitat diversity / structural variety increases the range of niches and hence species	5

Question	Marking guidance	Marks
04.2	<i>Any 3 from:</i> • Designated protected areas: SSSIs, NNRs (Sherwood Forest), SACs (Epping Forest), National Parks • Legal protection of ancient woodland: planning permission refused for developments destroying irreplaceable habitats (2012 NPPF) • Conservation management: coppicing, creating clearings, planting mixed native species, leaving dead wood, controlling deer • Planting new native woodlands linking fragments (corridors); Woodland Trust / National Trust ownership	3
04.3	• No species is completely dominant and food webs have many alternative links • So a change in the population of one species has a relatively small impact on the whole community	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-52; AQA spec 3.1.2.3.4.	
05.1	• Drainage makes soils aerobic and drier, increasing grass growth but wetland plants and the animals they support (waders, amphibians, invertebrates) cannot survive • Fertiliser increases nutrient availability so vigorous species (grasses, nettles) grow taller and out-compete smaller wildflowers for light; diversity falls • Hedgerow removal destroys habitat / corridors and nesting sites; loss of natural pest predator habitats increases pest damage; larger fields increase wind erosion; genetic contamination and introduced species also accepted	6
05.2	• Long-continued traditional farming has created plagioclimax habitats colonised by many species: hay meadows, chalk grassland, grazed heath / moorland, hedgerows • These communities depend on the farming continuing	2
05.3	• CO₂ from fossil fuel use in machinery and embodied energy of fertiliser (Haber process) and machinery; ploughing increases soil respiration • Methane from livestock intestines and rice paddies; nitrous oxide from manure and nitrogen fertilisers	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-330; AQA spec 3.5.1.3.	
06.1	• Ship rudders that turn in tighter circles / aerogenerator blades that start turning at lower wind speeds • Water-repelling surface of lotus flowers • Coatings for aircraft and ships that reduce fuel consumption / hospital coatings that bacteria cannot stick to • Hooked burrs on seeds that stick to animal fur	4
06.2	• The mound absorbs sunlight and warms; hot air inside rises, creating a convection current that draws stale air out of the lower parts (solar-driven ventilation) • Air blowing over the mound has a lower pressure which also draws air out • Copied in shopping complexes / office blocks to give natural ventilation and cooling without air conditioning, so less energy / electricity is used	3

Question	Marking guidance	Marks
06.3	<i>Any 2 from:</i> • Marsupials: young develop in the pouch so are easier to study than a human foetus, helping understanding of developmental problems • Hippopotamus skin secretes hipposudoric acid (natural sunscreen and antimicrobial) which may improve treatment of burns • Marine sponges produce proteins that prevent rejection of grafts, being developed to prevent rejection of transplanted organs	2
06.4	• They are light but strong: hollow with internal struts that prevent bending / breaking, copied in lightweight bridges and roofs	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 11-14; AQA spec 3.1.2.1.1.	
07.1	• About 40-50 traps • Because the running mean stops changing / fluctuating with fewer than this the mean is unreliable	2
07.2	• Widely scattered results (large standard deviation) mean a single sample may not be representative • More samples are needed so that the effect of variability is averaged out and the mean is representative / significance can be assessed	2
07.3	<i>Any 3 from:</i> • Same type and size of pitfall trap, set for the same length of time • Traps emptied at the same time of day / same weather conditions in both habitats • Same number of traps and same method of positioning (random or systematic) in each habitat • Other variables that could affect results (eg rainfall, temperature) monitored or controlled	3
07.4	• Flying insect activity changes with weather over minutes to weeks (temperature, wind) and diurnally • Samples on different occasions are needed to produce a mean that represents the typical value or to identify trends • The interval between samples should match the rate at which the factor changes, found from a preliminary study	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 391-393; AQA spec 3.7.1.	
08.1	• Yield increases as inputs increase • But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns)	2
08.2	• Second unit: $70 - 50 = 20$ units • Fifth unit: $94 - 90 = 4$ units	2
08.3	• The first units of input on each area give the largest yield increases • Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve	2
08.4	• Some areas have much greater potential productivity so investment there is sensible • Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons)	2
08.5	• Yield per unit area is very high because of large inputs • But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	
09.1	• In winter nutrients are high (storms disturb sediments, deep mixing) but light is too low for photosynthesis • In spring light rises while nutrients are still high so growth is rapid - the spring bloom (peak about May) • The bloom uses up nutrients (falling to 10 by July); dead plankton sink out of the photic zone carrying nutrients to the seabed • So nutrients become limiting in summer despite high light; a smaller autumn bloom follows when storms return nutrients while light is still adequate	4
09.2	• Warm surface water is stable (no seasonal mixing / storms) so nutrients lost to deep water in sinking dead organisms are not replaced • Nutrient (especially phosphate) concentrations in the photic zone remain very low - the limiting factor	2
09.3	• Upwelling (offshore winds, seamounts, currents deflected upwards) brings cold nutrient-rich deep water into the photic zone • Coastal waters receive nutrients washed off the land in rivers and storms disturb shallow sediments • Nutrients plus light in the shallow photic zone allow high algal productivity supporting rich food webs and fisheries (eg Peru anchovy)	3
09.4	• Phosphates have low solubility and no atmospheric reservoir; they are only added by river runoff or upwelling and are lost to sediments	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-339; AQA spec 3.5.2.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Context: wild catches have not increased since about 1990 as stocks are fully or over-exploited; aquaculture output growing rapidly and approaching the wild catch; fish are the last large-scale hunted food • Advantages of aquaculture: control of limiting factors, selective breeding, gender control, high yields per area; fish have low FCRs (no thermoregulation / support) compared with livestock; herbivores (tilapia, carp) and molluscs very efficient; polyculture, IMTA, aquaponics, rice-fish • Limits: carnivorous species (salmon, shrimp) need fishmeal from wild fish so still depend on fishing and deplete forage fish (sandeels - puffins); some species (bluefin tuna) will not breed in captivity; consumer preference for carnivores • Environmental impacts: habitat loss (mangroves), organic pollution, pesticides, antibiotics and resistance, lice, escapes and gene pool contamination, predator control; and how they can be reduced • Fishing can be sustainable with quotas, gear design, NTZs, size limits, effort limits; MSY management needs good data • Judgement: aquaculture can supply much of the growth in demand, especially herbivores and shellfish, but cannot fully replace fishing; both need sustainable management	9
10.2	• A species that is given artificial food (pellets) rather than relying on natural food in the water, eg salmon, shrimp	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-359; AQA spec 3.5.2.	

Question	Marking guidance	Marks
11.1	Natural systems have remained sustainable over very long periods. Discuss how the features of natural systems could be copied to make human systems more sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Dynamic equilibria and negative feedback in climate, hydrological cycle and populations; positive feedback and tipping points caused by humans • Energy: solar-driven, low energy density, low temperature (photosynthesis, enzymes, nitrogen fixation vs Haber process); renewable energy, heat pumps, bioleaching, fuel cells • Materials: cycles, few abundant elements, biodegradable non-toxic wastes vs fossil fuels and dispersed minerals; recycling, Cradle to Cradle, industrial symbiosis • Diversity and stability: rainforests / reefs vs agroecosystems; diverse energy mixes • Optimum vs maximum production; supporting ecosystems in agriculture, forestry, fisheries, urban design • Case studies and metrics; limits and judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-379; AQA spec 3.6.1-3.6.4.	25

Question	Marking guidance	Marks
11.2	Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Marine productivity: light, nutrients, upwelling; where fisheries occur • Methods: demersal / pelagic trawling, purse seining, drift nets, long lines, shellfish traps - effectiveness, selectivity, energy, impacts • Impacts: overfishing (r vs k selection, catchable vs breeding size), changed age structure, by-catch (albatross, porpoise, turtles, shrimp), ghost fishing, seabed / reef / seagrass damage, dynamite, food web effects (sandeels, tritons) • Population dynamics and MSY: Russell formula, difficulties of data, indicators from catches (CPUE, mean size, age), research on eggs and larvae • Management: quotas, mesh design, escape panels, TEDs, pingers, hook design, effort limits, method bans, NTZs, closed seasons, size limits, protected individuals, population seeding, biodegradable gear; EU CFP, IWC, CCAMLR • Evaluation with examples (cod, orange roughy, herring recovery, NTZ spillover); role of aquaculture; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 338-351; AQA spec 3.5.2.	25